Data Analytics and Insights Cheat Sheet

Key Fact

Did you know that 90% of the world's data has been created in the last two years? Yet, only a fraction of it is analyzed for decision-making.

Key Words and Formulas Cheat Sheet

Types of Data

- **Structured Data:** Organized data (e.g., tables, spreadsheets) that can be easily analyzed.
- **Unstructured Data:** Data without a predefined structure (e.g., videos, images, social media posts).
- **Qualitative Data:** Descriptive, non-numeric information (e.g., customer reviews).
- **Quantitative Data:** Numeric information that can be measured and analyzed statistically.

Data Cleaning Formulas

- **REMOVE DUPLICATES:**
  - **Google Sheets:** Navigate to **Data > Data Cleanup > Remove Duplicates**.
  - **Excel Formula:** `=UNIQUE(range)`
- **HANDLE MISSING VALUES:**
  - **Formula to Fill Missing Values:**
    - **IFERROR:** `=IFERROR(value, value_if_error)` to replace errors.
    - **AVERAGE:** `=IF(A1="", AVERAGE(B1:B10), A1)` to replace blanks with an average.
- **TRIM EXTRA SPACES:**
  - **Formula:** `=TRIM(text)`
- **STANDARDIZE DATA FORMATS:**
  - **LOWER:** `=LOWER(text)` (convert text to lowercase)
  - **UPPER:** `=UPPER(text)` (convert text to uppercase)
  - **PROPER:** `=PROPER(text)` (capitalize first letter of each word)

Data Analysis & Visualization

- **Choosing the Right Chart:**
  - **Bar Chart:** Compare values across categories.
  - **Line Chart:** Show trends over time.
  - **Pie Chart:** Display parts of a whole.
  - **Scatter Plot:** Visualize relationships between variables.
  - **Heatmap:** Highlight patterns or concentrations.
- **Data Visualization Tools in Google Sheets:**
  - **Insert Chart:** Go to **Insert > Chart** and customize data ranges.
  - **Dashboard Design:** Combine charts and slicers to create interactive dashboards.

Useful Formulas for Data Analysis

- **SUM:** Adds a range of numbers.
  - **Formula:** `=SUM(A1:A10)`
- **AVERAGE:** Calculates the mean.
  - **Formula:** `=AVERAGE(A1:A10)`
- **COUNT:** Counts numeric entries.
  - **Formula:** `=COUNT(A1:A10)`
- **COUNTA:** Counts all non-empty cells.
  - **Formula:** `=COUNTA(A1:A10)`
- **IF:** Performs conditional logic.
  - **Formula:** `=IF(condition, value_if_true, value_if_false)`
  - **Example:** `=IF(A1>100, "High", "Low")`
- **FILTER:** Returns rows that meet specific criteria.
  - **Formula:** `=FILTER(range, condition1, [condition2, ...])`
  - **Example:** `=FILTER(A1:A10, B1:B10>50)`
- **QUERY:** Perform SQL-like queries on data.
  - **Formula:** `=QUERY(data, query, [headers])`
  - **Example:** `=QUERY(A1:D100, "SELECT A, SUM(B) WHERE C > 100 GROUP BY A ORDER BY SUM(B) DESC")`

Data Cleaning & Transformation Tips

- **Use Conditional Formatting:** Highlight errors, duplicates, or trends.
  - Navigate to **Format > Conditional Formatting** and set rules (e.g., highlight cells greater than 100).

- **Split Text:** Break text into separate columns.
    - Use **Split Text to Columns** or the formula `=SPLIT(A1, " ")`.
- **Combine Data:** Use `=CONCATENATE(text1, text2)` or `=TEXTJOIN("-", TRUE, A1:A5)`.
